STAT/MA 41600

In-Class Problem Set #29: October 26, 2026

On these questions, you are encouraged to again **double-check your work**, by computing the expected values in two ways, namely, by using the joint probability density function, and by using the (single variable) probability density function for the variable in the particular question at hand. For instance, for $E(X^2)$, we can compute either: $E(X^2) = \int_{-\infty}^{\infty} x^2 \, f_X(x) \, dx$, or we can compute $E(X^2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^2 \, f_{X,Y}(x,y) \, dy \, dx$.

1. Suppose that X and Y are continuous random variables whose joint probability density function $f_{X,Y}(x,y)$ is constant on the interior of the triangle with vertices at $(0,0)$, $(8,0)$, and $(0,5)$, and otherwise $f_{X,Y}(x,y) = 0$.

1a. Find $\mathrm{Var}(X)$.

1b. Find $\mathrm{Var}(Y)$.

2. Suppose that X and Y have joint probability density function $f_{X,Y}(x,y) = 14e^{-2x-7y}$ for $x > 0$ and $y > 0$, and otherwise $f_{X,Y}(x,y) = 0$.

2a. Find $\mathrm{Var}(X)$.

2b. Find $\mathrm{Var}(Y)$.

3. Suppose that X and Y have joint probability density function

$$f_{X,Y}(x,y) = \begin{cases} \frac{4}{1875}(25 - xy) & \text{if } 0 < x < 5 \text{ and } 0 < y < 5, \\ 0 & \text{otherwise.} \end{cases}$$

Find $\mathrm{Var}(X)$. (By symmetry, $\mathrm{Var}(Y)$ will have the same value.)

4. Suppose that, for $0 < x < 5$ and $0 < y < 2$, we have $f_{X,Y}(x,y) = (1/25)(5-x)(2-y)$, and otherwise $f_{X,Y}(x,y) = 0$.

4a. Find $\mathrm{Var}(X)$.

4b. Find $\mathrm{Var}(Y)$.

5. Consider two random variables X and Y with joint probability density function $f_{X,Y}(x,y) = 63e^{-2x-7y}$ for $y > x > 0$, and otherwise $f_{X,Y}(x,y) = 0$.

5a. Find $\mathrm{Var}(X)$.

5b. Find $\mathrm{Var}(Y)$.